In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

In [ ]:
```
df = pd.read_csv("train.csv")
```

In [ ]:
```
df.shape
```

In [ ]:
```
df
```

In [ ]:
```
df.info()
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.isna().sum()
```

In [ ]:
```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

In [ ]:
```
df.isna().sum()
```

In [ ]:
```
def fun1(value):
    if (value == "male"):
        return 1
    else:
        return 0
```

In [ ]:
```
def fun1(value):
    if (value == "male"):
        return 1
    else:
        return 0
```

In [ ]:
```
def fun2(value):
    if (value == 'S'):
        return 0
    elif (value == 'C'):
        return 1
    elif (value == 'Q'):
        return 2
    else:
        return 0
```

In [ ]:
```
df["Sex"] = df["Sex"].apply(fun1)
```

In [ ]:
```
df["Embarked"] = df["Embarked"].apply(fun2)
```

In [ ]:
```
df = df.drop("Cabin", axis=1)
```

In [ ]:
```
df.shape
```

In [ ]:
```
px.box(df["Sex"], df["Age"], color=df["Survived"])
```

In [ ]:
```
plt.figure(figsize=(10,7))
box = sns.boxplot(df,x="Sex", y="Age", hue="Survived")
plt.show()
```

In [ ]:
```

```